To: Lena, Tomás **From:** Hana **Re:** First full run, all four cohorts (2014–2017) **For:** 30 June meeting

TL;DR. The first stage is there and it is strong. Internships move. Earnings are positive but only marginally significant ($p = 0.07$), so I would not headline them yet. Everything below uses cohort FE only. *No lottery-block FE yet* (see caveat 1). Code is in `/analysis/june_run/`, log file `main_0621.log`.

Definitions used in this run. Low-SES = Pell-eligible in the first year (our analysis sample). **High-SES = not Pell-eligible.**[1] Treatment = share of a student's floor-mates (excluding self) who are high-SES, scaled per 10 pp. A tie means co-swipes within 2 minutes on $\geq$10 distinct days in the semester, excluding the student's own hall entrance. Students need $\geq$5 ties to be in the network sample. EC_i = (share of i's ties who are high-SES)/0.5. With this definition the mean EC is above 1, because 70% of campus is non-Pell.

Table 1. Sample construction

	N	% of Pell sample
Entering first-years, 2014–2017	28,604	
Lived in University housing	26,118	
Assigned through standard lottery	24,871	
of which Pell-eligible (**analysis sample**)	**7,412**	100.0
Year-1 network sample ($\geq$5 ties)	6,463	87.2
Year-3 network sample	5,319	71.8
Consented to UI linkage	4,297	58.0
Positive in-state UI earnings at $\approx$25	3,412	46.0
Floors (floor $\times$ cohort)	628	

Table 2. First stage: effect on economic connectedness (per 10 pp high-SES floor-mates)

	(1) EC Yr 1	(2) EC Yr 1	(3) EC Yr 3
High-SES share	0.068***	0.066***	0.027**
	(0.012)	(0.011)	(0.013)
Controls	No	Yes	Yes
Mean of dep. var.	1.21	1.21	1.26
N	6,463	6,463	5,319

Cohort FE in all columns. SEs clustered by floor (628). Controls: female, first-gen, race, SAT, HS GPA, parental AGI, home-county EC. The survey-based EC (senior-survey friend nominations) is *not built yet*. I need to know whether our IRB approval covers linking the nominations to the aid records.

[1]The top-quintile AGI flag in the FAFSA extract is missing for about 18% of non-Pell students (no FAFSA filed), so for now I define high-SES as everyone who is not Pell-eligible. It's cleaner and has no missings. Mean share of non-Pell floor-mates = 0.702 (SD 0.083).

Table 3. Outcomes run so far (per 10 pp high-SES floor-mates)

Outcome	Mean	Coef.	s.e.	*N*
First-year GPA	3.02	0.012	(0.018)	7,412
Graduated within 4 years	0.61	0.005	(0.010)	7,412
Paid internship by junior year	0.37	0.020**	(0.009)	7,412
Internship at large employer	0.08	0.008	(0.005)	7,412
Employed at 25	0.83	0.006	(0.011)	4,297
Earnings at 25 ($)	38,420	2,080*	(1,150)	3,412
Log earnings at 25	10.44	0.051*	(0.029)	3,412
Employed at high-wage firm	0.21	0.017*	(0.009)	4,297

Cohort FE + controls; SEs clustered by floor. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The other four outcomes in our list (cumulative GPA, STEM/econ major, business/econ major, 6-year graduation, grad school) are coded but not yet run. That's five, actually, so the list is 13, not 12. Earnings are in 2024 $.

Caveats / things I'm worried about

1. **No block FE.** The lottery runs within 48 blocks (first-choice residential area × honors × cohort). I asked IR for the block file on 14 May and followed up on 2 June, but haven't heard back. Until we have it we can't include block FE. My guess is the first stage survives, but I really don't know about earnings.
2. **Consent timing.** UI consent is Q41 on the *senior exit survey*. So the earnings sample = reached senior year *and* answered the survey *and* said yes. That's post-treatment. Treatment → missing earnings: −0.020 (0.013). Not significant, but it's not zero either.
3. **Network attrition.** 12.8% of Pell students have no Year-1 network, compared with 6.3% of non-Pell. Treatment → missing network: −0.017 (0.011).
4. **Proximity.** 38% of ties are floor-mates and another 24% are same residential area but a different floor. I think that's mostly the dining commons, since fall meal plans only work in your own area's commons. So 62% of "friendships" are people you'd bump into anyway. I haven't figured out what to do about this.
5. UI data are in-state only. People who moved out of state show up as "not employed" or missing.

Next steps (mine): block FE once IR responds; build survey EC once the IRB question is answered; run the remaining outcomes; heterogeneity (first-gen, race, home-county EC) if we want it for the brown bag.